

Topic # 6B – Cost & EVA
Chapter 7 – PMBOK



Cost Control & Earned Value Management

- Performance review meetings can be a powerful tool to help control project costs
- Knowing you have to report on your progress is an incentive for people to perform better
- Performance measurement is another important tool for cost control
  - There are many general accounting approaches for measuring cost performance but ***earned value management*** (EVM) is a unique tool to project management
- Earned Value Management is also known as Earned Value Analysis(EVA)

Earned Value Management (EVM)

- **EVM** is a project performance measurement technique that integrates scope, time, and cost data
- Given a **baseline** (original plan plus approved changes), you can determine how well the project is meeting its goals
- You must enter actual information periodically to use EVM
  - Was a WBS item completed or approximately how much of the work was completed
  - Actual start and end dates
  - Actual cost
- More and more organizations around the world are using EVM to help control project costs

- The **planned value (PV),** formerly called the budgeted cost of work scheduled (BCWS), also called the budget, is that portion of the approved total cost estimate planned to be spent on an activity during a given period
- **Actual cost (AC),** formerly called actual cost of work performed (ACWP), is the total of direct and indirect costs incurred in accomplishing work on an activity during a given period
- The **earned value (EV),** formerly called the budgeted cost of work performed (BCWP), is an estimate of the value of the physical work actually completed
  - EV is based on the original planned costs for the project or activity and the rate at which the team is completing work on the project or activity to date

Earned Value Calculation

Earned Value Calculations for one activity after one week

Activity	Week 1	Week 2	Total	% Complete after week 1	EV after week 1
Purchase & Install Web Server	10,000	0	10,000	75%	7,500
Planned Value (PV)	10,000				
Actual Cost (AC)	15,000	5000	20,000		
Cost Variance (CV)	-7500				
Schedule Variance (SV)	-2500				
Cost Performance Index (CPI)	50%				
Schedule Performance Index (SPI)	75%				

EVM

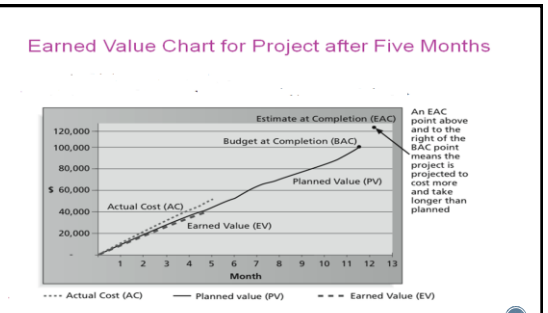
- The **rate of performance (RP)** is the ratio of actual work completed to the percentage of work planned to have been completed at any given time during the life of the project or activity.
- **Cost variance (CV)** is the earned value minus the actual cost.
- **Schedule variance (SV)** is the earned value minus the planned value.
- The **cost performance index (CPI)** is the ratio of earned value to actual cost and can be used to estimate the projected cost of completing the project.
- The **schedule performance index (SPI)** is the ratio of earned value to planned value and can be used to estimate the projected time to complete the project.
- The cost performance index can be used to calculate the **estimate at completion (EAC)** an estimate of what it will cost to complete the project based on performance to date.

Earned Value Formulas

Term	Formula
Earned Value	EV = PV to date × RP
Cost Variance	CV = EV – AC
Schedule Variance	SV = EV – PV
Cost Performance Index	CPI = EV/AC
Schedule Performance Index	SPI = EV/PV
Estimate at Completion (EAC)	EAC = BAC/CPI
Estimated Time to Complete	Original Time Estimate/SPI

Earned Value Calculations for one year Project after 5 months

Activity	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec	PV	% Complete	EV
Plan & Staff Project	4,000	4000											8,000	100	8000
Analyze Requirements		6000	6000										12,000	100	12000
Develop ERDs			4000	4000									8,000	100	8000
Develop Database tables				6000	4000								10,000	100	10000
Design forms, reports & queries					8000	4000							12,000	50	6000
Build Prototype						10000							10,000		
Evaluate Prototype						2000	6000						8,000		
Changes as per feedback							4000	6000	4000				14,000		
Test System									4000	4000	2000		10,000		
Document System											3000	1000	4,000		
Train Users												4000	4,000		
Monthly Planned Value(PV)	4,000	10,000	10,000	10,000	12,000	16,000	10,000	6,000	8,000	4,000	5,000	5,000	100,000		
Commulative Planned Value(PV)	4,000	14,000	24,000	34,000	46,000	62,000	72,000	78,000	86,000	90,000	95,000	100,000			
Monthly Actual Cost (AC)	4000	11000	11000	12000	15000										
Commulative Actual Cost (AC)	4000	15000	26000	38000	53000										
Monthly Earned Value (EV)	4000	10000	10000	10000	10000										
Commulative Earned Value(EV)	4000	14000	24000	34000	44000										
Project EV on May 31	44000														
Project PV on May 31	46000														
Project AC on May 31	53000														
SV	-9000			Estimate at Completion (EAC)			120454.55								
SV	-2000			Estimated time to complete			12.545455								
CPI	83.0188679														
SPI	95.65217391														



END OF TOPIC 6

- **COMING UP!!!!!!**
- **Project Team**
- **Resource Allocation**
- **Project Control**
- **Midterm Exam II**
- **Agile Software Project Management**